

BIOLOGY (THEORY) – MOCK TEST 2

Statistically Repeated Board Questions Checklist

Time: 3 Hours

Max Marks: 70

By Pragya Thunderstudy

SECTION A (16 Marks)

1. Which one of the following hormones is secreted by the human placenta that helps in the maintenance of pregnancy?

- | | |
|--------------|---------------|
| (A) Relaxin | (B) hCG |
| (C) Oxytocin | (D) Prolactin |

[Source: 97, 100]

2. The histone core in a nucleosome of chromatin thread is an:

- | | |
|--------------|--------------|
| (A) Octamer | (B) Hexamer |
| (C) Tetramer | (D) Pentamer |

[Source: 6]

3. Restriction Endonuclease – Hind II cuts DNA molecules by recognizing a specific sequence of:

- | | |
|---------------------|----------------------|
| (A) Six base pairs | (B) Four base pairs |
| (C) Five base pairs | (D) Three base pairs |

[Source: 103, 105]

4. An organism used as a biofertilizer for raising soybean crops is:

- | | |
|------------------|-----------------|
| (A) Azospirillum | (B) Rhizobium |
| (C) Nostoc | (D) Azotobacter |

[Source: 97]

5. Interferons are proteins secreted by:

- | | |
|-----------------------------|--------------------------|
| (A) RBCs | (B) WBCs |
| (C) Bacteria-infected cells | (D) Virus-infected cells |

[Source: 45, 81]

6. Which of the following is not an example of aneuploidy?

- | | |
|-----------------------|---------------------|
| (A) Turner's syndrome | (B) Down's syndrome |
| (C) Phenylketonuria | (D) Klinefelter's |

[Source: 6]

7. The milk colostrum, produced during initial lactation, is rich in which antibody?

- (A) IgE (B) IgA
(C) IgG (D) IgM

[Source: 6, 82]

8. Identify the correct statement regarding PCR:

- (A) Thermostable DNA polymerase used (B) Primers not required
(C) DNA fragment separation (D) Inside living cell only

[Source: 79]

9. In a transcription unit, the promoter is located towards:

- (A) 5' structural gene (B) 3' structural gene
(C) 5' coding strand (D) 3' coding strand

[Source: 6]

10. Organ transplant immunosuppressive agent molecule is produced by:

- (A) T. polysporum (B) M. purpureus
(C) S. cerevisiae (D) P. notatum

[Source: 8, 56]

11. Which of the following is a palindromic sequence?

- (A) 5'-GAATTC-3' (B) 5'-GGGACC-3'
(C) 5'-CCTAGG-3' (D) 5'-ATCGCTA-3'

[Source: 9, 53]

12. Technique used to detect HIV infection in suspected AIDS patients:

- (A) PCR (B) ELISA
(C) MRI (D) Widal Test

[Source: 9, 83]

Q13-16: Assertion (A) & Reason (R)

13. (A): In birds, sex is determined by males. (R): Males are ZZ, females are ZW. [Source: 51]
14. (A): Amniocentesis is often misused. (R): Used for sex determination female foeticide. [Source: 12]
15. (A): Agrobacterium is a dicot pathogen. (R): Delivers T-DNA to form tumors. [Source: 107]
16. (A): High-species communities are more stable. (R): Result in less biomass variation. [Source: 51]

SECTION B (10 Marks)

17. Interaction between **Fig tree and Wasp**. Why is it Mutualism? [Source: 14, 136]

18. Convention for naming **EcoRI**. Write its recognized palindromic sequence. [Source: 14, 118]

19. Name two Gonadotropins and explain their role in **Spermatogenesis**. [Source: 21]

20. (a) What are "**Flocs**" in sewage treatment? (b) Primary vs Activated Sludge difference. [Source: 46, 129]

21. List any two female VDs and two complications if left untreated. [Source: 12]

SECTION C (21 Marks)

22. Mode of action of **Oral Pills** and **Copper-T IUDs**. [Source: 10, 114]

23. Describe **S.L. Miller experiment**. Name scientists whose hypothesis stimulated him. Conclusion? [Source: 114]

24. DNA: 5'-ATG AGG GGG ATG-3'. (a) mRNA formed (b) Amino acids coded (c) Deletion of 'A' effect. [Source: 86, 135]

25. Pyramid of Biomass (Sea): (a) Upright or inverted? Why? (b) Name producers/consumers. [Source: 16, 113]

26. (a) Define **Transgenic Animals**. (b) Role in Vaccine Safety and Bios (Rosie). [Source: 19, 60, 118]

27. Processing of **hnRNA** to **mRNA** (Splicing, Capping, Tailing). [Source: 16, 58]

28. Syndrome: Short stature, furrowed tongue, open mouth. (a) Identify (b) Cause (c) Autosomal/Sex-linked? [Source: 15, 45]

SECTION D (8 Marks)

29. Case Study: Menstrual Cycle. [Source: 62, 122]

(a) Identify 'Ovulatory phase' and hormone surge. (b) Endometrium in Luteal phase and hormone. (c) Why does menstruation occur?

30. Case Study: Cloning Vectors (pBR322). [Source: 16, 65, 121]

(a) Selectable markers function (Amp^R , Tet^R). (b) BamHI insert resistance inactivation. (c) Blue-White screening principle.

SECTION E (15 Marks)

31. (a) Male Gametophyte development with diagrams. (b) Anatropous ovule structure.

OR

(a) Sectional view of Human Ovary follicles stages. (b) Oogenesis events. [Source: 16, 92, 124]

32. (a) Hershey and Chase experiment steps (Infection, Blending, Centrifugation) and Aim.

OR

(b) Meselson and Stahl experiment for semi-conservative replication. [Source: 16, 71]

33. (a) STP biological treatment involving BOD, Floccs, Activated Sludge.

OR

(b) Bt Cotton formation: Cry gene toxin kill mechanism and inactive vs active state. [Source: 16, 29, 130]

**Biology Mock Test 2 By Pragya Thunderstudy
Success through Strategy – Pink Series**